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(54) **HEAT ENGINES, SYSTEMS FOR PROVIDING PRESSURIZED REFRIGERANT, AND RELATED METHODS**

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See application file for complete search history.

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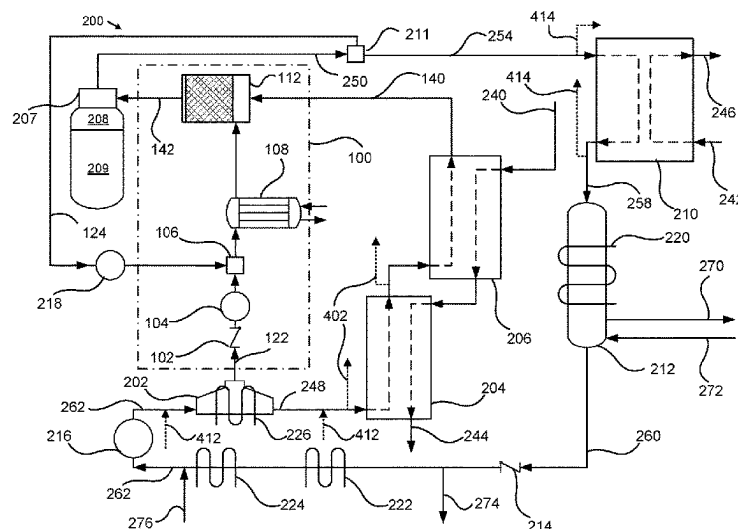
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(57) **ABSTRACT**

A method for generating power from a heat source includes mixing a refrigerant in a liquid phase with a lubricating oil, heating the mixture to evaporate the refrigerant, mixing the heated mixture with additional refrigerant in a superheated phase, and atomizing the lubricating oil to disperse the lubricating oil within the refrigerant. The atomized lubricating oil and the refrigerant are passed through a decompressor to generate an electrical current. The refrigerant may be an organic material having a boiling point below about -35°C . Related systems and heat engines are also disclosed.

18 Claims, 3 Drawing Sheets



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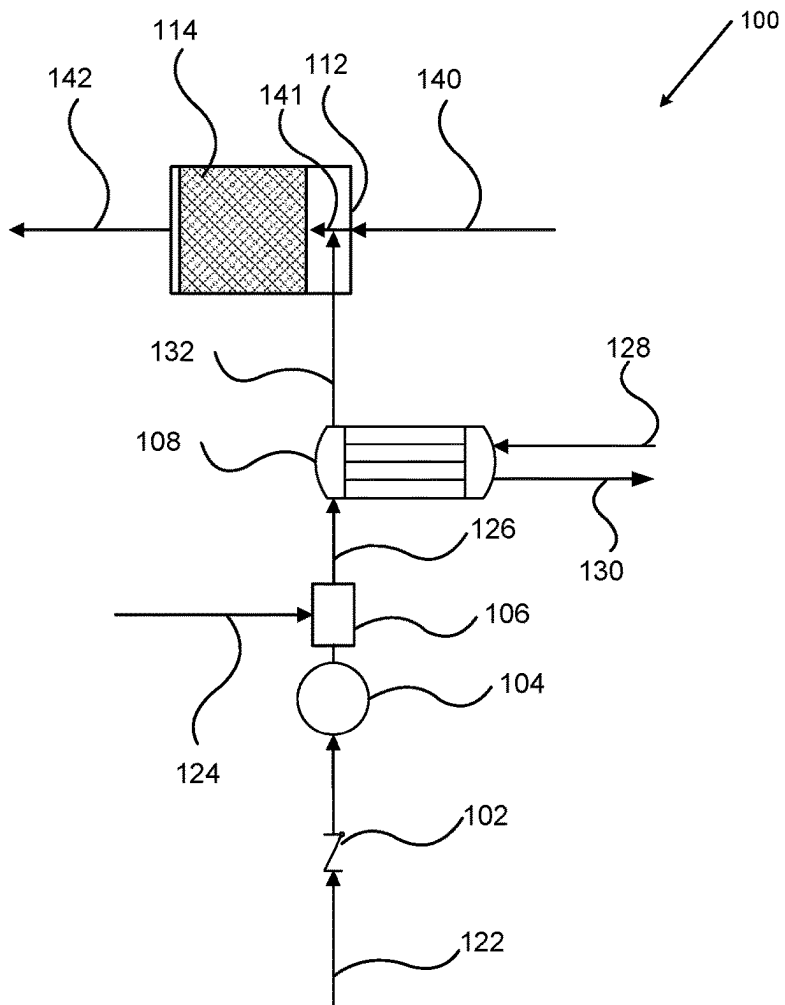


FIG. 1

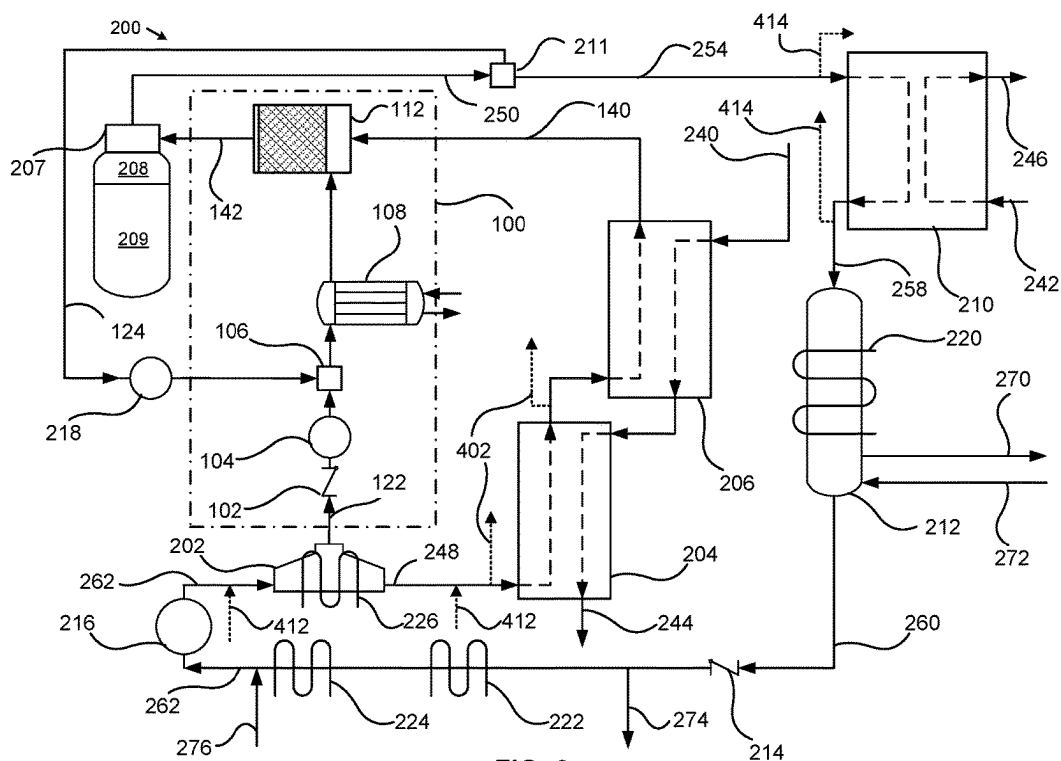


FIG. 2

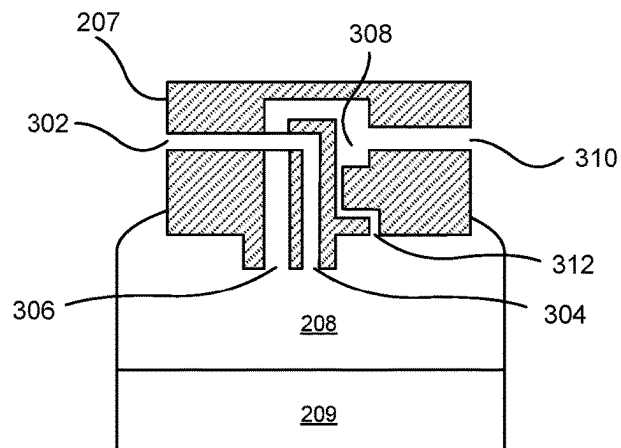


FIG. 3

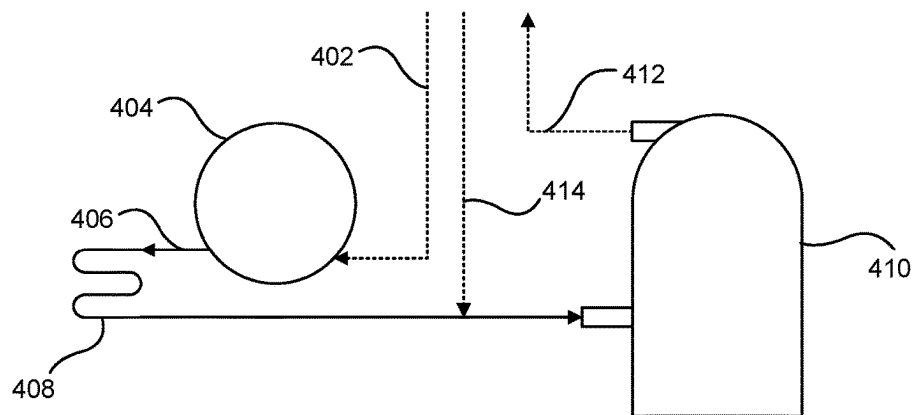


FIG. 4

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HEAT ENGINES, SYSTEMS FOR PROVIDING PRESSURIZED REFRIGERANT, AND RELATED METHODS

Matter enclosed in heavy brackets [] appears in the original patent but forms no part of this reissue specification; matter printed in italics indicates the additions made by reissue; a claim printed with strikethrough indicates that the claim was canceled, disclaimed, or held invalid by a prior post-patent action or proceeding.

NOTICE: This is a reissue continuation of U.S. Patent Reissue application Ser. No. 17/351,489, filed Jun. 18, 2021, now U.S. Pat. No. RE 49,730, issued Nov. 21, 2023, which is a broadening application for reissue of U.S. Pat. No. 10,323,545 B2, issued Jun. 18, 2019, to Johnson for Heat Engines, Systems for Providing Pressurized Refrigerant, and Related Methods. Because the present application is a reissue application, the amendments are made pursuant to 37 C.F.R. § 1.173, which requires the use of brackets and underlining.

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation reissue of Reissue application Ser. No. 17/351,489 (now Reissue U.S. Pat. No. RE 49,730), filed Jun. 18, 2021, which is a reissue of U.S. patent application Ser. No. 15/576,566 (now U.S. Pat. No. 10,323,545), which is a national phase entry on Nov. 22, 2017 under 35 U.S.C. § 371 of International Patent Application No. PCT/US2016/034178, filed May 25, 2016, [designating the United States of America and published in English as International Patent Publication WO 2016/196144 A1 on Dec. 8, 2016,] which claims the benefit [under Article 8 of the Patent Cooperation Treaty to] of U.S. Provisional Patent Application [Ser.] No. 62/169,971, filed [Jun.] Jun. 2, 2015.

TECHNICAL FIELD

Embodiments of the present disclosure relate generally to systems and methods for power generation and other processes involving compressed refrigerants. For example, the systems and methods may be used for generating electricity from a heat source.

BACKGROUND

The Rankine cycle is a thermodynamic cycle in which heat is converted into work. The heat is supplied externally to a closed loop, typically with water as the working fluid. The Rankine cycle generates about 80% of all electric power used throughout the world, and is used by solar thermal, biomass, coal and nuclear power plants. Rankine-cycle power systems typically transform thermal energy into electrical energy. A conventional Rankine cycle power system employs the following four basic steps: (1) thermal energy is used, in a boiler, to turn water into steam; (2) the steam is sent through a turbine, which, in turn, drives an electric generator; (3) the steam is condensed back into water by discharging the remaining thermal energy in the steam to the environment; and (4) the condensate is pumped back to the boiler. In an ideal Rankine cycle, the expansion is isentropic (i.e., at constant entropy) and the evaporation and conden-

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sation processes are isobaric (i.e., at constant pressure). However, irreversibilities in real processes lower cycle efficiency. Those irreversibilities are primarily attributable to two factors: conversion of some energy into heat during expansion of the working fluid during step (2) of the cycle; and inefficiency caused by pressure drops in the heat exchangers during steps (1) and (3).

The efficiency of a Rankine cycle is a function of the physical properties of the working fluid. Without the pressure reaching supercritical levels for the working fluid, the temperature range over which the cycle can operate is quite small. For example, conventional turbine entry temperature limits are around 565° C. (the creep limit of stainless steel) and condenser temperatures are around 30° C. This gives a theoretical Carnot efficiency of about 63% compared with an actual efficiency of 42% for a modern coal-fired power station. Low turbine entry temperature (compared with an internal-combustion gas turbine) is why the Rankine cycle is often used as a bottoming cycle in combined-cycle gas turbine power stations. The working fluid in a Rankine cycle follows a closed loop and is re-used continually. While many working fluids can and have been used in the Rankine cycle, water is usually the fluid of choice because it is abundant, inexpensive, nontoxic, generally non-reactive, and possesses favorable thermodynamic properties. Organic Rankine cycles (ORCs) have been developed to enable recovery of energy from lower temperature sources, such as industrial waste heat, geothermal heat, solar ponds, etc. Working fluids in ORCs are organic, high molecular mass fluids having a liquid-vapor phase change (i.e., boiling point) at a lower temperature than the water-steam phase change at a given pressure. Using an ORC, heat from lower temperature sources can be converted to useful work that can be harnessed to generate electricity.

ORC technology can be used to recover energy from waste heat. For example, the technology can be applied to heat and power plants, industrial and farming processes (e.g., organic products fermentation), hot exhausts from ovens or furnaces, flue gas condensation, exhaust gases from vehicles, intercooling of a compressor, and the condenser of a power cycle. ORC technology can also be used to extract useful energy from biomass, geothermal heat sources, solar fields, etc.

ORCs are described in International Patent Publication No. WO 2014/124061, titled "Improved Organic Rankine Cycle Decompression Heat Engine," published Aug. 14, 2014, the entire disclosure of which is hereby incorporated herein by this reference.

BRIEF SUMMARY

In some embodiments, a method includes mixing a lubricating oil with a first portion of a refrigerant in a liquid phase to form a mixture and heating the mixture of the lubricating oil and the first portion of the refrigerant to form a heated mixture, wherein at least a portion of the first portion of the refrigerant is in a gaseous phase. The method includes mixing the heated mixture with a second portion of the refrigerant in a superheated phase and atomizing the lubricating oil to disperse the lubricating oil within the refrigerant. The refrigerant may exhibit a boiling point below about -35° C. In some embodiments, the refrigerant may include an organic material.

In other embodiments, a heat engine includes a high-pressure zone configured to transfer heat from at least one heat source to a refrigerant and configured to contain a first portion of the refrigerant in a gaseous phase; a low-pressure

zone configured to transfer heat from the refrigerant to at least one heat sink and configured to contain a second portion of the refrigerant in a liquid phase; a bypass zone configured to mix a third portion of the refrigerant in a liquid phase with a lubricating oil; and an atomizer comprising a mixing media configured to atomize the lubricating oil and disperse the lubricating oil within the first and third portions of the refrigerant. The refrigerant exhibits a boiling point below about -35°C . A substantially closed-loop path for the refrigerant comprises the high-pressure zone, the low-pressure zone, and the bypass zone.

In some embodiments, a system for providing a pressurized refrigerant includes a pump configured to move a first portion of a refrigerant in a liquid phase through a conduit; a mixing device for mixing a lubricating oil with the first portion of the refrigerant in the liquid phase; at least one heat source configured to transfer heat to the first portion of the refrigerant in the liquid phase; another mixing device for mixing a second portion of the refrigerant in a gaseous phase with the lubricating oil and the first portion of the refrigerant; and a mixing media configured to atomize the lubricating oil and disperse the lubricating oil within the refrigerant. The refrigerant exhibits a boiling point below about -35°C . The at least one heat source is configured to evaporate the first portion of the refrigerant.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified schematic block diagram illustrating a system for providing a pressurized refrigerant, according to the present disclosure;

FIG. 2 is a simplified schematic block diagram illustrating a heat engine including the system shown in FIG. 1;

FIG. 3 is a simplified diagram illustrating a manifold that may be used in the heat engine illustrated in FIG. 2; and

FIG. 4 is a simplified schematic block diagram illustrating how refrigerant in the heat engine shown in FIG. 2 may be used in conjunction with other equipment.

DETAILED DESCRIPTION

The following description provides specific details, such as material types, compositions, material thicknesses, and processing conditions in order to provide a thorough description of embodiments of the disclosure. However, a person of ordinary skill in the art will understand that embodiments of the disclosure may be practiced without employing these specific details. Indeed, embodiments of the disclosure may be practiced in conjunction with conventional techniques employed in the industry. In addition, the description provided below does not form a complete process flow for providing pressurized fluids or operating heat engines. Only those process acts and structures necessary to understand the embodiments of the disclosure are described in detail below. A person of ordinary skill in the art will understand that some process components (e.g., pipes, line filters, valves, temperature detectors, flow detectors, pressure detectors, etc.) are inherently disclosed herein and that adding various conventional process components and acts would be in accord with the disclosure. Additional acts or materials may be provided by adding conventional elements to those disclosed.

As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise.

As used herein, the term "and/or" includes any and all combinations of one or more of the listed items.

As used herein, relational terms, such as "first," "second," "top," "bottom," "upper," "lower," "over," "under," etc., are used for clarity and convenience in understanding the disclosure and accompanying drawings and do not connote or depend on any specific preference, orientation, or order, except where the context clearly indicates otherwise.

As used herein, the term "substantially," in reference to a given parameter, property, or condition, means to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances.

FIG. 1 is a simplified schematic block diagram illustrating a system **100** for providing a pressurized refrigerant. The system **100** may be used to provide pressurized refrigerant to a heat engine, a refrigeration system, a heat pump, or any other system in which pressurized refrigerant is used. In some embodiments, the system **100** may provide pressurized refrigerant to a heat engine using an organic Rankine cycle (ORC).

As shown in FIG. 1, the system **100** may include a check valve **102** configured to enable flow of refrigerant in only one direction. A pump **104** may be configured to receive refrigerant from the check valve **102** and provide a force to drive refrigerant through the system **100**. The pump **104** may be, for example a variable speed circulation pump. The pump **104** may be significantly smaller than pumps conventionally used for providing pressurized refrigerants because the pump **104** need not significantly increase the pressure of the refrigerant. Instead, the refrigerant pressure may be increased by heating, as explained in further detail below.

The refrigerant may be an organic material having a boiling point below about -35°C . Such a refrigerant may be used to extract energy from heat sources having relatively low temperatures (e.g., below about 82°C .) and convert that energy into electricity more efficiently than refrigerants in conventional systems. For example, the refrigerant may be R-410A, a near-azeotropic mixture of difluoromethane and pentafluoroethane, which has a boiling point of approximately -51°C . Other refrigerants may also be used, such as those known in the art as R-22, R-502, R-507, R-13, R-503, R-23, R-717 (ammonia), or R-744 (carbon dioxide (CO_2)). Such refrigerants are commercially available, for example, from Airgas Refrigerants, Inc., of Lawrenceville, Ga., USA.

A mixing device **106** may be configured to mix lubricating oil with the organic refrigerant after the organic refrigerant passes through the pump **104**. The mixing device **106** may include, for example, a mixing tee, a mix chamber, an aspirator, or any other means for mixing fluids. The lubricating oil may be an organic material formulated to remain in a liquid phase at the operating conditions of a refrigeration process, and to have beneficial lubrication and stability properties. For example, the lubricating oil may include mineral oil (e.g., naphthalenic) or synthetic oils (e.g., glycols, esters, alkylbenzenes, polyolesters, etc.). In some embodiments, the lubricating oil may also include a lubricity enhancer, such as an additive package sold under the trade name A/C Re-New by Nu-Calgon, of St. Louis, Mo., USA. The lubricity enhancer may be formulated to have a higher lubricity than the lubricating oil.

The system **100** may further include one or more heat sources **108** depicted as a heat exchanger in FIG. 1. The one or more heat sources **108** may include condensers, heating coils, or any other heat source. The one or more heat sources **108** may include at least one wall through which heat may be transferred to the organic refrigerant and the lubricating oil. Though one heat source **108** is depicted in FIG. 1, the

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system **100** may include any number of heat sources, such as one, two, three, four, etc. The one or more heat sources **108** may be configured to evaporate the organic refrigerant to form a gas. For example, the one or more heat sources **108** may include an evaporator or a superheater.

The system **100** may further include an atomizer **112** configured to mix the organic refrigerant and lubricating oil with another portion of organic refrigerant, and to atomize the lubricating oil. The atomizer **112** may include a body having mixing and atomizing means **114** therein. For example, the atomizer **112** may include mixing vanes, a metal mesh (e.g., a sintered stainless steel mesh, such as a 316L stainless steel mesh material having a mesh size of about 1,000 microns, sold by Dorstener Wire Tech, of Spring, Tex., USA, under the name SINTERPORE®), a packing media, or any other structure for mixing and/or atomizing. In some embodiments, the atomizer **112** may further include a filter media (which may be replaceable) or other means to prevent unatomized liquid oil from leaving the atomizer **112**. The atomizer **112** may be configured to operate with minimal restriction, such that the pressure drop within the atomizer **112** is small or negligible.

The system **100** may be used to provide a pressurized, heated refrigerant. A refrigerant **122** may pass through the check valve **102** to the pump **104**, which pumps the refrigerant **122** through the system **100**. In some embodiments, the pump **104** may provide the sole means for circulating the refrigerant **122**. The refrigerant **122** leaving the pump **104** may enter the mixing device **106** and mix with a lubricating oil **124** to form a mixture **126**. The mixture **126** may then be heated in the heat source **108**. The heat source **108** may receive a hot fluid **128** having a temperature of less than about 80° C., less than about 70° C., less than about 60° C., or even less than about 50° C. Heat from the hot fluid **128** is transferred to the mixture **126**, such that a cooled fluid **130** (containing material that entered the heat source **108** as the hot fluid **128**) and a heated mixture **132** (containing material that entered the heat source **108** as the mixture **126**) leave the heat source **108**. The volumetric flow rate of the heated mixture **132** may be greater than the volumetric flow rate of the mixture **126** due to evaporation of the refrigerant **122**.

The mixture **126** may, in some embodiments, be exposed to additional heat sources. Transferring heat from the heat source **108** to the mixture **126** may involve increasing the specific volume of the mixture **126**.

The heat source **108** may include, for example, a waste heat source, an exhaust gas, a compressor intercooler, biomass, a geothermal heat source, a solar array, or any other heat source. The heat source **108** need not be at a temperature as high as those required in conventional systems, so the system **100** may be used to extract energy from heat sources that are traditionally not used for power generation.

The heated mixture **132** (which includes the lubricating oil **124** mixed with the refrigerant **122**) and another portion of gaseous or superheated refrigerant **140** may be mixed within the atomizer **112** to form a superheated mixture **141**. Alternatively, the heated mixture **132** and the refrigerant **140** may be mixed before entering the atomizer **112** to form the superheated mixture **141**, such as in a mixing tee, a mix chamber, an aspirator, or any other means for mixing fluids. The refrigerant **140** may typically have the same composition as the refrigerant **122**. The refrigerant **140** may be at or near its critical temperature. Within the atomizer **112**, the lubricating oil may become atomized to form a fine mist, which may be dispersed within the refrigerant. For example, the superheated mixture **141** may be passed through a metal mesh. In some embodiments, the lubricating oil **124** may be

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approximately uniformly dispersed throughout the refrigerant. The refrigerant and the lubricating oil may leave the atomizer **112** as an atomized mixture **142**. In some embodiments, the atomized mixture **142** may be filtered or otherwise processed to enhance homogeneity of the atomized mixture **142**.

In some embodiments, the atomized mixture **142** (i.e., the pressurized refrigerant and the atomized lubricating oil) may be used in an ORC heat engine. For example, FIG. 2 is a simplified schematic block diagram illustrating a heat engine **200** including the system **100** shown in FIG. 1 (note that the reference numerals for some material flows shown in FIG. 1 are omitted from FIG. 2 for simplicity). The heat engine **200** generally includes a high-pressure zone, a low-pressure zone, and a bypass zone. When the heat engine **200** is in operation, the pressure of the working refrigerant within the high-pressure zone is approximately constant, with variations in temperature along the flow path. Similarly, within the low-pressure zone, the pressure of the working refrigerant is approximately constant (but at a lower pressure than in the high-pressure zone). The refrigerant in the bypass zone is at approximately the same pressure as in the high-pressure zone. Because the high-pressure zone and the low-pressure zone are each at approximately constant pressures, refrigerant may be passed through one or more external loops if additional heat transfer is desired or required, as discussed below and shown in FIG. 3.

The heat engine **200** includes a holding tank **202** for containing refrigerant. Pressurized refrigerant **248** travels from the holding tank **202** to a heat source **204** (e.g., an evaporator) and a superheater **206**, which may be generally referred to as heat sources **204**, **206**. The heat sources **204**, **206** may each include at least one wall through which heat may be transferred to the refrigerant **248** from a hot fluid **240**. Though two heat sources **204**, **206** are depicted in FIG. 2, the heat engine **200** may include any number of heat sources, such as one, two, three, four, etc. One of the heat sources **204**, **206** may be configured to evaporate the pressurized refrigerant **248** to form a gas, which may correspond to the refrigerant **140** shown in FIG. 1. After the refrigerant **140** leaves the superheater **206**, the refrigerant **140** may be at or near its critical temperature.

The heat sources **204**, **206** may receive the hot fluid **240** having a temperature of less than about 80° C., less than about 70° C., less than about 60° C., less than about 50° C., or even less than about 40° C. Heat from the hot fluid **240** is transferred to the refrigerant **248**. The hot fluid **240** may leave the heat sources **204**, **206** as fluid **244** for disposal or other uses.

The refrigerant **140** may then be mixed with the heated mixture **132** of refrigerant and oil that has been preheated in the heat source **108** (see also FIG. 1). Within the atomizer **112**, the lubricating oil **124** may become atomized to form a fine mist, which may be dispersed within the refrigerant **140** as the atomized mixture **142**, as discussed above and shown in more detail in FIG. 1.

The atomized mixture **142** may then enter a manifold **207** having an expansion device **208** coupled to and configured to drive an electrical generator **209**. The expansion device **208** may include a positive displacement decompressor configured to provide a pressure gradient through which the organic refrigerant in the gaseous phase flows continuously from the high-pressure zone to the low-pressure zone. The expansion device **208** may be configured to maintain a pressure differential between the high-pressure zone and the low-pressure zone of between about 10 bar and about 100 bar, and may be configured to extract mechanical energy due

to the pressure gradient. For example, the expansion device **208** may be configured to maintain a pressure differential between the high-pressure zone and the low-pressure zone of between about 20 bar and about 42 bar.

In some embodiments, the expansion device **208** may be a turbine, a decompressor, a twin-screw expander, or any other appropriate device configured to rotate a portion of the electrical generator **209**. For example, the expansion device **208** may include an orbital scroll decompressor. The expansion device **208** may include materials formulated to withstand operating conditions. For example, the expansion device **208** may include stainless steel. A mixture **250** leaving the expansion device **208** may be at a relatively lower pressure and a relatively higher temperature than the atomized mixture **142** entering the expansion device **208**. Thus, the high-pressure zone of the heat engine **200** may be defined to include the holding tank **202**, the heat sources **204**, **206**, the atomizer **112**, and the expansion device **208**.

The manifold **207** may be configured to receive the atomized mixture **142** from the atomizer **112** and to provide the mixture **250** at a lower pressure and higher temperature than the atomized mixture **142**. The manifold **207** may be configured to use heat from the atomized mixture **142** to assist in separation of oil from the mixture **250**. The mixture **250** may rise vertically within the manifold **207** through orifices or a steel mesh. This may promote separation of oil, which tends to be heavier, from refrigerant. The oil may tend to fall back down into the electrical generator **209** below, where it may be reused for sealing and lubrication.

As shown in FIG. 3, the manifold **207** may include a refrigerant inlet **302** through which the atomized mixture **142** (FIG. 2) may be received. The atomized mixture **142** may pass into the expansion device **208** via a protrusion **304**. Refrigerant leaving the expansion device **208** (i.e., as mixture **250**, shown in FIG. 2) enters the manifold **207** through an opening **306**. The refrigerant travels through a separation region **308**, where oil is removed from the refrigerant to form the mixture **250** (FIG. 2), which leaves the manifold **207** via a refrigerant outlet **310**. The oil removed may return to the expansion device **208** and/or electrical generator **209** via an oil drain **312**.

Combining the refrigerant inlet **302** and the refrigerant outlet **310** may simplify installation of the heat engine **200**, and may provide a means to use some heat from the incoming atomized mixture **142** to assist in oil separation. The manifold **207** may include flow paths configured to promote heat transfer between incoming and outgoing refrigerant.

Furthermore, the manifold **207** may include one or more tortuous paths through which the refrigerant may pass before leaving the manifold **207**. For example, the opening **306** may include one or a plurality of bored holes or orifices, which may optionally be packed with a steel mesh or other material having a high surface area. Forcing the exhaust vapor to rise vertically through such a path may enhance separation of oil from the refrigerant (e.g., because the oil may be heavier than the refrigerant).

The exhaust vapor may rise vertically within the manifold **207**, then flow around sharp edges, before flowing downward again, thus promoting the separation of heavier oil, which may drain downward from the separation region **308** through the opening **306** back to the expansion device **208**, based on gravity.

The protrusion **304** may be configured such that the atomized mixture **142** entering the expansion device **208** does not disrupt the flow of the atomized mixture **142** via the separation region **308**. For example the protrusion **304** may

extend downward from the center of the manifold **207** so as not to allow the atomized mixture **142** entering the expansion device **208** to easily force any mixed oil leaving the expansion device **208** to enter the opening **306**. That is, refrigerant leaving the expansion device **208** may enter the manifold **207** through the opening **306** via the protrusion **304**, and may retain some oil. Such an arrangement may also promote heat transfer between incoming and outgoing refrigerant, which may enhance oil separation.

The electrical generator **209** may be any device configured to convert mechanical energy extracted from the refrigerant in the expansion device **208** to electrical energy, such as an alternating-current generator having a rotor and a stator. The expansion device **208** may be located vertically above or below the electrical generator **209**. In some embodiments, the expansion device **208** and the electrical generator **209** may be integrated into a single device.

At least a portion of the lubricating oil **124** may be extracted from refrigerant **254** in a separator **211**, which may be, for example, a trap, a filter, etc. The lubricating oil **124** may be returned to the refrigerant **122** via a pump **218** and the mixing device **106**. The low-pressure, hot refrigerant **254** may enter a condenser **210**, where it may be cooled by a heat sink **242** to condense the refrigerant **254**, forming a condensed refrigerant **258**. The heat sink **242** may be cold water (e.g., culinary water, well water, etc.). The heat sink **242** may be at approximately ambient or room temperature. For example, the heat sink **242** may enter the condenser **210** at a temperature from about 5° C. to about 35° C., such as from about 15° C. to about 25° C. The heat sink **246** leaving the condenser **210** may be from about 1° C. to about 15° C. warmer than the heat sink **242**, which temperature difference may depend on the flow rates of the heat sink **242** and refrigerant **254**, as well as the temperatures thereof. The refrigerant **254** may be partially or fully condensed in the condenser **210**.

The condensed refrigerant **258** may be transferred to a liquid refrigerant tank **212**, where the condensed refrigerant **258** may optionally be further cooled by one or more additional heat sink **220** to form cooled refrigerant **260**. The heat sink **220** may be, for example, cold water, a thermal expansion valve (TXV), a thermoelectric device, or any other material or device capable of absorbing or transferring heat. The heat engine **200** may be operable to use multiple heat sinks **220** of different types and at different temperatures. Heat sinks **220** may be added to balance heat loads with other processes. In some embodiments, the condensed refrigerant **258** or a portion thereof may be cycled through a conduit **270** out of the heat engine **200** to an external heat sink for cooling, and subsequently returned to the heat engine **200** through another conduit **272** (e.g., back to the liquid refrigerant tank **212**). In such embodiments, the refrigerant may be considered as having a substantially closed-loop path that includes the heat engine **200** and the external heat sink.

A pressure control system **214** may regulate the flow and pressure of the cooled refrigerant **260** from the liquid refrigerant tank **212**. The pressure control system **214** may include a check valve, a pressure regulator, a dampening chamber, a manual or electronic controller, and/or any other selected control device.

After passing through the pressure control system **214**, heat sources **222**, **224** may optionally preheat the cooled refrigerant **260** to form a preheated refrigerant **262**. The heat sources **222**, **224** may be, for example, a waste heat source, exhaust gas, a compressor intercooler, biomass, a geothermal heat source, a solar array, or any other material or device

capable of generating or transferring heat. In some embodiments, the cooled refrigerant **260** or a portion thereof may be cycled through a conduit **274** out of the heat engine **200** to an external heat source for heating, either instead of or in addition to the heat sources **222**, **224**, and subsequently returned to the heat engine **200** through another conduit **276**. In such embodiments, the refrigerant may be considered as having a substantially closed-loop path that includes the heat engine **200** and the external heat source. Thus, the heat engine **200** may be operable to use multiple heat sources **222**, **224** of different types and at different temperatures. Heat sources **222**, **224** may be added to balance heat loads with other processes.

A circulation pump **216** may return the preheated refrigerant **262** to the holding tank **202**, and may separate the low-pressure zone from the high-pressure zone. Thus, the low-pressure zone of the heat engine **200** may be defined to include the separator **211**, the condenser **210**, the liquid refrigerant tank **212**, and the pressure control system **214**. The circulation pump **216** may provide continuous flow of the preheated refrigerant **262** in the liquid phase from the low-pressure zone to the high-pressure zone. The circulation pump **216** may be, for example, a positive displacement hydraulic pump. The circulation pump **216** may be configured to operate as a liquid pressure booster pump, increasing the pressure of the preheated refrigerant **262** as it moves the preheated refrigerant **262** through the heat engine **200**. Pumping the preheated refrigerant **262** in the liquid phase may be a relatively more efficient way to increase the pressure than pumping a gas-phase refrigerant. The holding tank **202** may optionally be further preheated by another heat source **226**. The heat source **226** may be, for example, a waste heat source, exhaust gas, a compressor intercooler, biomass, a geothermal heat source, a solar array, or any other material or device capable of generating or transferring heat. The use of multiple heat sources **222**, **224**, **226** to preheat the refrigerant (in addition to the heat sources **204**, **206**) may increase the amount of energy generation by increasing the total amount of heat transferred to the refrigerant in the cycle.

The heat engine **200** includes a bypass zone, in which a relatively small amount of the refrigerant **122** is mixed with lubricating oil **124**. The bypass zone may be defined to include the check valve **102**, the pump **104**, the mixing device **106**, and the heat source **108**.

The lubricating oil **124** may help to seal the expansion device **208** to limit or prevent slippage of the moving parts of the expansion device **208**. The lubricating oil **124** may also increase the density of the refrigerant entering the expansion device **208** as the atomized mixture **142**. However, if lubricating oil **124** passes through the expansion device **208** in bulk form, the lubricating oil **124** may stall the expansion device **208**, such as by blocking flow of the refrigerant. Thus, in the bypass zone, the lubricating oil **124** may be pre-mixed with a portion of the refrigerant **122** to improve dispersion. The operation of units in the bypass zone (in particular, the check valve **102**, the pump **104**, the mixing device **106**, and the heat source **108**) are described above with respect to FIG. 1. The atomizer **112** may further promote dispersion of the lubricating oil **124** in the refrigerant, such that the lubricating oil **124** can effectively lubricate the moving parts of the expansion device **208** without inhibiting flow of the refrigerant. Thus, the lubricating oil **124** may increase the efficiency of the heat engine **200** (by reducing friction losses in the expansion device **208**) and increase the expected operating life thereof.

The heat engine **200** may include a means for mixing the refrigerant **122** and lubricating oil **124** from the bypass zone with the refrigerant **140** from the superheater **206** before these materials enter the atomizer **112**, such as in a mixing tee, a mix chamber, an aspirator, or any other means for mixing fluids. In some embodiments, these materials may be mixed within the atomizer **112** or within a body also containing the atomizer **112**.

The atomized mixture **142** entering the expansion device **208** may be from about 0.01% to about 10% by mass (i.e., a ratio of a mass flow rate of the refrigerant **122**, **140** to a mass flow rate of the lubricating oil **124** may be between about 10:1 and about 10,000:1). For example, the atomized mixture **142** may be from about 0.1% to about 1.0% by mass (i.e., a ratio of the mass flow rate of the refrigerant **122**, **140** to the mass flow rate of the lubricating oil **124** may be between about 100:1 and about 1000:1). The atomized mixture **142** may be about 5.0% by mass (i.e., a ratio of the mass flow rate of the refrigerant **122**, **140** to the mass flow rate of the lubricating oil **124** may be about 20:1).

A method for generating power from a heat source includes mixing a portion of the refrigerant in a liquid phase with a lubricating oil, heating the mixture to evaporate the portion of the refrigerant, mixing the heated mixture with a superheated portion of the refrigerant, atomizing the lubricating oil, and passing the atomized lubricating oil and the refrigerant through a decompressor coupled with an electrical generator to generate an electrical current. At least a portion of the lubricating oil may then be separated from the refrigerant, and the refrigerant may be condensed back to liquid form.

The power may be generated without using a compressor, and thus, the power output may be higher than power output from a conventional heat engine with the same inputs. That is, the heat engine may be more efficient in power conversion than conventional systems.

In the heat engine **200**, heat flows into the refrigerant in the heat sources **204**, **206**. The heat is converted to mechanical work, and to electricity in the expansion device **208** and electrical generator **209**. Waste heat is removed from the heat engine **200** via the condenser **210**. Minor amounts of energy are consumed pumping the refrigerant, lubricating oil, and heat sources and sinks (e.g., water), but the heat engine **200** does not require a compressor to produce pressurized refrigerant **140** (FIG. 1). Thus, the heat engine **200** may operate more efficiently than conventional heat engines.

Refrigerant may be extracted from the heat engine **200**, used in another device or operation, and returned to the heat engine **200**. For example, and as shown in FIG. 4, a refrigerant **402** in liquid form may enter a thermal expansion device **404**. The refrigerant **402** may be converted to a gaseous refrigerant **406**. A coil **408** may be used as a heat sink for refrigeration or air conditioning.

The gaseous refrigerant **406** may enter a variable-speed compressor **410**, which may produce a pressurized hot gaseous refrigerant **412** suitable for return to the heat engine **200** (FIG. 2). Another gaseous refrigerant **414** may also be compressed in the variable-speed compressor **410**, instead of or in addition to the gaseous refrigerant **406**. The amount of energy required to operate the variable-speed compressor **410** such that the pressurized hot gaseous refrigerant **412** may be returned to the heat engine **200** may be less than the energy required to operate conventional cooling equipment.

Returning to FIG. 2, the refrigerant **402** in liquid form may be separated from the refrigerant **248** (either before or after the heat source **204**), or any other selected point in the heat engine **200**. The refrigerant **414** in gaseous form may be

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separated from the refrigerant **254**, the refrigerant **258**, or any other selected point in the heat engine **200**. The pressurized hot gaseous refrigerant **412** may be returned to the heat engine **200** after the circulation pump **216**, after the holding tank **202**, or at any other appropriate location. Because the heat engine **200** has high-pressure zone and a low-pressure zone, the extraction or addition point for refrigerants may be at any point within each zone. That is, by matching the pressure, the refrigerant may flow and mix with refrigerant in the heat engine **200** without the need for additional conditioning.

EXAMPLES

Prophetic Example 1: Mathematical Modeled Heat Engine

A heat engine similar to the heat engine **200** shown in FIG. **2** is expected to produce 9.88 kW of electricity from a hot water source at 69.0° C., with a cold water source at 17.0° C., using R-410A refrigerant. The expansion device **208** is an orbital scroll, a commercially available scroll compressor designed for air conditioning, model SH380, available from Danfoss A/S, of Nordborg, Denmark, operating at 1517 RPM. The compressor is operated in the reverse orientation of its intended use—thus it operates as an expander rather than as a compressor.

Calculated pressures and temperatures for the refrigerant and the hot and cold water supplies are shown in Table 1.

TABLE 1

Operating Pressures and Temperatures		
	Temperature (° C.)	Pressure (bar)
Evaporator hot water, inlet	69.0	
Evaporator hot water, outlet	63.0	
Condenser cooling water, inlet	17.0	
Condenser cooling water, outlet	21.0	
Refrigerant at circulation pump outlet	23.0	36.0
Refrigerant at evaporator inlet	23.0	36.0
Refrigerant at superheater outlet	67.0	36.0
Refrigerant at scroll inlet	67.0	36.0
Refrigerant at scroll outlet	21.5	15.6
Refrigerant at condenser inlet	21.5	15.0
Refrigerant at condenser outlet	21.0	15.0
Refrigerant at circulation pump inlet	21.0	15.0

Additional operating parameters are shown in Table 2.

TABLE 2

Other Operating Parameters	
Refrigerant mass flow	304.5 g/s
Hot water flow	282 l/min
Cold water flow	482 l/min
Heat transfer from hot water to refrigerant	113.4 kW
Heat transfer from refrigerant to cold water	112.0 kW
Carnot efficiency (theoretical maximum)	15.20%
Maximum work output	17.23 kW
Gross work output from scroll	11.14 kW
Gross efficiency	9.83%
Gross percentage of Carnot efficiency	64.7%
Power consumed by pumps (oil pump, refrigerant pump, and water pumps)	1.27 kW
Net power produced	9.88 kW
Net efficiency	8.71%
Net percentage of Carnot efficiency	57.3%

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Example 2: Heat Engine Operating at 1,500 RPM

A heat engine similar to the heat engine **200** shown in FIG. **2** was operated to produce about 11 kW of electricity from a hot water source at about 68.9° C., with a cold water source at about 16.3° C., using R-410A refrigerant. The expansion device **208** was an orbital scroll, a commercially available scroll compressor designed for air conditioning, model SH380, from Danfoss A/S, operating at about 1,500 RPM. Selected operating parameters as observed are shown in Table 3.

TABLE 3

Observed Operating Parameters	
Evaporator hot water, inlet	68.9° C.
Evaporator hot water, outlet	61.8° C.
Hot water flow	260 l/min
Condenser cooling water, inlet	16.3° C.
Condenser cooling water, outlet	12.7° C.
Cold water flow	485 l/min
Carnot efficiency (theoretical maximum)	15.3%
Net power produced	11 kW
Net efficiency	8.8%
Net percentage of Carnot efficiency	57.5%

Example 3: Heat Engine Operating at 1121 RPM

A heat engine similar to the heat engine **2** shown in FIG. **200** was operated to produce about 6 kW of electricity from a hot water source at about 58.1° C., with a cold water source at about 15.5° C., using R-410A refrigerant. The expansion device **208** was an orbital scroll, a commercially available scroll compressor designed for air conditioning, model SH380, available from Danfoss A/S, operating at about 1,121 RPM. Selected operating parameters as observed are shown in Table 4.

TABLE 4

Observed Operating Parameters	
Evaporator hot water, inlet	58.1° C.
Evaporator hot water, outlet	52.2° C.
Hot water flow	191 l/min
Condenser cooling water, inlet	15.5° C.
Condenser cooling water, outlet	12.4° C.
Cold water flow	376 l/min
Carnot efficiency (theoretical maximum)	12.9%
Net power produced	6 kW
Net efficiency	7.1%
Net percentage of Carnot efficiency	55.0%

Additional non limiting example embodiments of the disclosure are described below.

Embodiment 1

A method includes mixing a lubricating oil with a first portion of a refrigerant in a liquid phase to form a mixture and heating the mixture of the lubricating oil and the first portion of the refrigerant to form a heated mixture, wherein at least a portion of the first portion of the refrigerant is in a gaseous phase. The method includes mixing the heated mixture with a second portion of the refrigerant in a superheated phase and atomizing the lubricating oil to disperse the lubricating oil within the refrigerant.

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Embodiment 2

The method of Embodiment 1, wherein atomizing the lubricating oil comprises passing the lubricating oil and the refrigerant through a metal mesh.

Embodiment 3

The method of Embodiment 1 or Embodiment 2, further comprising pumping the first portion of the refrigerant through a conduit before mixing the lubricating oil therewith.

Embodiment 4

The method of any of Embodiments 1 through 3, wherein heating the mixture of the lubricating oil and the first portion of the refrigerant to form a heated mixture comprises transferring heat to the mixture from at least one source selected from the group consisting of a waste heat source, an exhaust gas, a compressor intercooler, biomass, a geothermal heat source, and a solar array.

Embodiment 5

The method of any of Embodiments 1 through 4, wherein the refrigerant exhibits a boiling point below about -35°C .

Embodiment 6

The method of any of Embodiments 1 through 5, further comprising: passing the atomized lubricating oil and the refrigerant through a decompressor operatively associated with an electrical generator to decrease a pressure of the refrigerant and generate an electrical current; separating at least a portion of the lubricating oil from the refrigerant; and condensing at least a portion of the refrigerant to re-form the first portion of the refrigerant.

Embodiment 7

The method of Embodiment 6, wherein condensing at least a portion of the refrigerant to form the first portion of the refrigerant comprises transferring heat from the refrigerant to each of a first heat sink and a second heat sink.

Embodiment 8

The method of any of Embodiments 1 through 7, wherein heating the mixture of the lubricating oil and the first portion of the refrigerant to form a heated mixture comprises increasing a specific volume of the mixture.

Embodiment 9

The method of any of Embodiments 1 through 8, further comprising passing the atomized lubricating oil and the refrigerant through a replaceable filter.

Embodiment 10

The method of any of Embodiments 1 through 9, wherein atomizing the lubricating oil comprises maintaining the lubricating oil and the refrigerant at approximately constant pressure.

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Embodiment 11

The method of any of Embodiments 1 through 10, further comprising mixing an additive with the mixture, the additive exhibiting a higher lubricity than the lubricating oil.

Embodiment 12

The method of any of Embodiments 1 through 11, wherein mixing a lubricating oil with a first portion of a refrigerant to form a mixture comprises mixing the first portion of the refrigerant in a liquid phase with the lubricating oil downstream of a pump circulating the first portion of the refrigerant.

Embodiment 13

A heat engine comprising a high-pressure zone configured to transfer heat from at least one heat source to a refrigerant and configured to contain a first portion of the refrigerant in a gaseous phase; a low-pressure zone configured to transfer heat from the refrigerant to at least one heat sink and configured to contain a second portion of the refrigerant in a liquid phase; a bypass zone configured to mix a third portion of the refrigerant in a liquid phase with a lubricating oil; and an atomizer comprising a mixing media configured to atomize the lubricating oil and disperse the lubricating oil within the first and third portions of the refrigerant. The refrigerant exhibits a boiling point below about -35°C . A substantially closed-loop path for the refrigerant comprises the high-pressure zone, the low-pressure zone, and the bypass zone.

Embodiment 14

The heat engine of Embodiment 13, wherein the heat engine is configured to circulate the refrigerant in a Rankine cycle.

Embodiment 15

The heat engine of Embodiment 13 or Embodiment 14, further comprising a mixing device configured to mix the lubricating oil with the third portion of the refrigerant in the liquid phase.

Embodiment 16

The heat engine of any of Embodiments 13 through 15, wherein the high-pressure zone comprises at least one wall of the at least one heat source configured to transfer heat from the at least one heat source to the first portion of the refrigerant in the liquid phase, wherein the at least one heat source is configured to evaporate the first portion of the refrigerant.

Embodiment 17

The heat engine of any of Embodiments 13 through 16, further comprising a positive displacement decompressor configured to provide a pressure gradient through which the refrigerant in the gaseous phase flows continuously from the high-pressure zone to the low-pressure zone, the positive displacement decompressor configured to maintain a pressure differential between the high-pressure zone and the low-pressure zone of between about 20 bar and about 42 bar,

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the positive displacement decompressor configured to extract mechanical energy due to the pressure gradient.

Embodiment 18

The heat engine of Embodiment 17, further comprising an electrical generator coupled to the positive displacement decompressor and configured to convert the extracted mechanical energy to electrical energy.

Embodiment 19

The heat engine of Embodiment 18, further comprising a positive displacement hydraulic pump for providing continuous flow of the refrigerant in the liquid phase from the low-pressure zone to the high-pressure zone.

Embodiment 20

The heat engine of any of Embodiments 13 through 19, wherein the at least one heat source comprises a material having a temperature of less than about 60° C.

Embodiment 21

The heat engine of any of Embodiments 13 through 20, wherein the refrigerant circulates through the closed-loop path absorbing additional heat without passing through a compressor.

Embodiment 22

The heat engine of any of Embodiments 13 through 21, wherein the refrigerant absorbs or rejects heat while circulating through the closed-loop path without passing through a compressor.

Embodiment 23

The heat engine of any of Embodiments 13 through 22, wherein a ratio of a mass flow rate of a sum of the first and third portions of the refrigerant to a mass flow rate of the lubricating oil is between about 10:1 and about 100:1.

Embodiment 24

The heat engine of any of Embodiments 13 through 23, wherein the substantially closed-loop path further comprises at least one heat-transfer conduit external to the heat engine.

Embodiment 25

A system for providing a pressurized refrigerant comprises a pump configured to move a first portion of a refrigerant in a liquid phase through a conduit; a mixing device for mixing a lubricating oil with the first portion of the refrigerant in the liquid phase; at least one heat source configured to transfer heat to the first portion of the refrigerant in the liquid phase; another mixing device for mixing a second portion of the refrigerant in a gaseous phase with the lubricating oil and the first portion of the refrigerant; and a mixing media configured to atomize the lubricating oil and disperse the lubricating oil within the refrigerant. The refrigerant exhibits a boiling point below about -35° C. The at least one heat source is configured to evaporate the first portion of the refrigerant.

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Embodiment 26

The system of Embodiment 25, wherein the mixing media comprises metal mesh.

Embodiment 27

The method, system, or heat engine of any of Embodiments 1 through 26, wherein the refrigerant comprises an organic refrigerant.

Embodiment 28

The method, system, or heat engine of any of Embodiments 1 through 26, wherein the refrigerant comprises ammonia.

Embodiment 29

The method, system, or heat engine of any of Embodiments 1 through 26, wherein the refrigerant comprises carbon dioxide.

While the present disclosure has been described herein with respect to certain illustrated embodiments, those of ordinary skill in the art will recognize and appreciate that it is not so limited. Rather, many additions, deletions, and modifications to the illustrated embodiments may be made without departing from the scope of the invention as hereinafter claimed, including legal equivalents thereof. In addition, features from one embodiment may be combined with features of another embodiment while still being encompassed within the scope of the disclosure as contemplated by the inventors. Further, embodiments of the disclosure have utility with different and various heat engine types.

What is claimed is:

[1. A method comprising:

mixing a lubricating oil with a first portion of a refrigerant in a bypass zone of a heat engine to form a mixture, the first portion of the refrigerant in a liquid phase;

heating the mixture of the lubricating oil and the first portion of the refrigerant in a high-pressure zone of the heat engine to form a heated mixture, wherein at least a portion of the first portion of the refrigerant is in a gaseous phase, wherein the refrigerant exhibits a boiling point below -35° C.;

mixing the heated mixture with a second portion of the refrigerant, the second portion of the refrigerant in a superheated phase;

atomizing the lubricating oil in an atomizer comprising a mixing media to disperse the lubricating oil within the refrigerant; and

cooling the refrigerant in a low-pressure zone of the heat engine through at least one heat sink;

wherein the high-pressure zone, the low-pressure zone, and the bypass zone form a closed-loop path for the refrigerant.]

[2. The method of claim 1, wherein atomizing the lubricating oil comprises passing the lubricating oil and the refrigerant through a metal mesh.]

[3. The method of claim 1, wherein heating the mixture of the lubricating oil and the first portion of the refrigerant to form a heated mixture comprises transferring heat to the mixture from at least one source selected from the group consisting of a waste heat source, an exhaust gas, a compressor intercooler, biomass, a geothermal heat source, and a solar array.]

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[4. The method of claim 1, further comprising:
passing the atomized lubricating oil and the refrigerant
through a decompressor operatively associated with an
electrical generator to decrease a pressure of the refrig-
erant and generate an electrical current;
separating at least a portion of the lubricating oil from the
refrigerant; and
condensing at least a portion of the refrigerant to re-form
the first portion of the refrigerant.]

[5. The method of claim 4, wherein the at least one heat
sink comprises a first heat sink and a second heat sink, and
wherein condensing at least a portion of the refrigerant to
form the first portion of the refrigerant comprises transfer-
ring heat from the refrigerant to each of the first heat sink
and the second heat sink.]

[6. The method of claim 1, wherein heating the mixture of
the lubricating oil and the first portion of the refrigerant to
form a heated mixture comprises increasing a specific vol-
ume of the mixture.]

[7. The method of claim 1, wherein atomizing the lubri-
cating oil comprises maintaining the lubricating oil and the
refrigerant at constant pressure.]

[8. The method of claim 1, further comprising mixing an
additive with the mixture, the additive exhibiting a higher
lubricity than the lubricating oil.]

[9. The method of claim 1, wherein mixing a lubricating
oil with a first portion of a refrigerant to form a mixture
comprises mixing the first portion of the refrigerant in a
liquid phase with the lubricating oil downstream of a pump
circulating the first portion of the refrigerant.]

[10. A heat engine, comprising:

a high-pressure zone configured to transfer heat from at
least one heat source to a refrigerant and configured to
contain a first portion of the refrigerant in a gaseous
phase, the refrigerant exhibiting a boiling point below
−35° C.;

a low-pressure zone configured to transfer heat from the
refrigerant to at least one heat sink and configured to
contain a second portion of the refrigerant in a liquid
phase;

a bypass zone configured to mix a third portion of the
refrigerant with a lubricating oil, the third portion of the
refrigerant and the lubricating oil in a liquid phase; and
an atomizer comprising a mixing media configured to
atomize the lubricating oil and disperse the lubricating
oil within the first and third portions of the refrigerant;
wherein a closed-loop path for the refrigerant comprises
the high-pressure zone, the low-pressure zone, and the
bypass zone.]

[11. The heat engine of claim 10, wherein the heat engine
is configured to circulate the refrigerant in a Rankine cycle.]

[12. The heat engine of claim 10, further comprising a
mixing device configured to mix the lubricating oil with the
third portion of the refrigerant in the liquid phase.]

[13. The heat engine of claim 10, wherein the high-
pressure zone comprises at least one wall of the at least one
heat source configured to transfer heat from the at least one
heat source to the first portion of the refrigerant in the liquid
phase, wherein the at least one heat source is configured to
evaporate the first portion of the refrigerant.]

[14. The heat engine of claim 10, further comprising a
positive displacement decompressor configured to provide a
pressure gradient through which the refrigerant in the gas-
eous phase flows continuously from the high-pressure zone
to the low-pressure zone, the positive displacement decom-
pressor configured to maintain a pressure differential
between the high-pressure zone and the low-pressure zone

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of between 20 bar and 42 bar, the positive displacement
decompressor configured to extract mechanical energy due
to the pressure gradient.]

[15. The heat engine of claim 14, further comprising an
electrical generator coupled to the positive displacement
decompressor and configured to convert the extracted
mechanical energy to electrical energy.]

[16. The heat engine of claim 15, further comprising a
positive displacement hydraulic pump for providing con-
tinuous flow of the refrigerant in the liquid phase from the
low-pressure zone to the high-pressure zone.]

[17. The heat engine of claim 10, wherein the closed-loop
path further comprises at least one heat-transfer conduit
external to the heat engine.]

18. A method comprising:

*mixing a lubricating oil with a first portion of a refrig-
erant in a bypass zone of a heat engine to form a
mixture, wherein the first portion of the refrigerant is in
a liquid phase;*

*heating the mixture of the lubricating oil and the first
portion of the refrigerant in a high-pressure zone of the
heat engine to form a heated mixture, wherein at least
a portion of the first portion of the refrigerant is in a
gaseous phase;*

*mixing the heated mixture with a second portion of the
refrigerant, wherein the second portion of the refrig-
erant is in a gaseous or superheated phase;*

*atomizing the lubricating oil in an atomizer holding a
mixing media to disperse the lubricating oil within the
refrigerant; and*

*cooling the refrigerant in a low-pressure zone of the heat
engine through at least one heat sink,
wherein the high-pressure zone, the low-pressure zone,
and the bypass zone form a closed-loop path for the
refrigerant.*

19. A method comprising:

*mixing a lubricating oil with a first portion of a refrig-
erant in a bypass zone to form a mixture;*

*heating the mixture of the lubricating oil and the first
portion of the refrigerant to form a heated mixture,
wherein at least a portion of the first portion of the
refrigerant is in a gaseous phase, and wherein the
lubricating oil and the refrigerant are mixed in a
mixing device to form the mixture;*

*mixing the heated mixture with a second portion of the
refrigerant, wherein the second portion of the refrig-
erant is in a gaseous or superheated phase;*

*atomizing the lubricating oil in an atomizer holding a
mixing media to disperse the lubricating oil within the
refrigerant; and*

cooling the refrigerant through at least one heat sink.

20. The method of claim 19, wherein atomizing the
lubricating oil includes passing the lubricating oil and the
refrigerant through a metal mesh.

21. The method of claim 19, wherein heating the mixture
of the lubricating oil and the first portion of the refrigerant
to form the heated mixture includes transferring heat to the
mixture from at least one source selected from the group
consisting of a waste heat source, an exhaust gas, a com-
pressor intercooler, biomass, a geothermal heat source, and
a solar array.

22. The method of claim 19, further comprising:

*passing the atomized lubricating oil and the refrigerant
through a decompressor operatively associated with an
electrical generator to decrease a pressure of the
refrigerant and generate an electrical current;*

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separating at least a portion of the lubricating oil from the refrigerant; and
condensing at least a portion of the refrigerant to a liquid phase.

23. The method of claim 22, wherein:

the at least one heat sink includes a first heat sink and a second heat sink, and

condensing the portion of the refrigerant to the liquid phase includes transferring heat from the refrigerant to each of the first heat sink and the second heat sink.

24. The method of claim 19, wherein heating the mixture of the lubricating oil and the refrigerant to form the heated mixture includes increasing a specific volume of the mixture.

25. The method of claim 19, wherein atomizing the lubricating oil includes maintaining the lubricating oil and the refrigerant at constant pressure.

26. The method of claim 19, further comprising mixing an additive with the mixture, wherein the additive exhibits a higher lubricity than the lubricating oil.

27. The method of claim 19, further comprising mixing the lubricating oil with the first portion of the refrigerant in a liquid phase in the mixing device to form the mixture of the lubricating oil and the first portion of the refrigerant.

28. A heat engine, comprising:

a high-pressure zone configured to transfer heat from at least one heat source to a refrigerant and configured to contain a first portion of the refrigerant in a gaseous phase;

a low-pressure zone configured to transfer heat from the refrigerant to at least one heat sink and configured to contain a second portion of the refrigerant in a liquid phase,

a bypass zone configured to mix a third portion of the refrigerant with a lubricating oil, the third portion of the refrigerant and the lubricating oil in a liquid phase; and

an atomizer holding a mixing media configured to atomize the lubricating oil and disperse the lubricating oil within the first portion of the refrigerant,

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wherein a closed-loop path for the refrigerant includes the high-pressure zone, the low-pressure zone, and the bypass zone.

29. The heat engine of claim 28, wherein the heat engine is configured to circulate the refrigerant in a Rankine cycle.

30. The heat engine of claim 28, further comprising a mixing device configured to mix the lubricating oil with the third portion of the refrigerant in the liquid phase in the bypass zone.

31. The heat engine of claim 28, wherein:

the high-pressure zone includes at least one wall of the at least one heat source configured to transfer heat from the at least one heat source to the first portion of the refrigerant in the liquid phase, and
the at least one heat source is configured to evaporate the first portion of the refrigerant.

32. The heat engine of claim 28, further comprising:

a positive displacement decompressor configured to provide a pressure gradient through which the refrigerant in the gaseous phase flows continuously from the high-pressure zone to the low-pressure zone,

wherein the positive displacement decompressor is configured to maintain a pressure differential between the high-pressure zone and the low-pressure zone of between 20 bar and 42 bar, and

wherein the positive displacement decompressor is configured to extract mechanical energy due to the pressure gradient.

33. The heat engine of claim 32, further comprising an electrical generator coupled to the positive displacement decompressor and configured to convert the extracted mechanical energy to electrical energy.

34. The heat engine of claim 33, further comprising a positive displacement hydraulic pump configured to provide continuous flow of the refrigerant in the liquid phase from the low-pressure zone to the high-pressure zone.

35. The heat engine of claim 28, wherein the closed-loop path includes at least one heat-transfer conduit external to the heat engine.

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